

▼ Introduction to Classification(Machine learning)

What is Classification ?

- Classification is a supervised learning technique in Machine Learning where the goal is to predict a category or label for a given input data.

Key Idea

- Instead of predicting a continuous value (like salary or house price in regression), classification predicts discrete outputs (classes).

Examples of classification Problems

- Yes / No
- Spam / Not spam.
- Pass/Fail
- Disease / not Disease
- Cat/ Dog classification.
- Approved / Rejected loan.

1. Logistic Regression

- Logistic Regression is a supervised learning algorithm used for classification problems.
- It is mainly used when the output variable is binary (two classes), such as
 - Yes/No
 - Pass/Fail
 - 0/1
 - Spam/Not spam

- Spam/Not spam

- In Logistic Regression , does not predict continuous values.
- Instead , it predicts the probability of a class.

Example:

- If we want to predict whether a student will pass:
 - Input : study hours , attendance , marks
 - Output: probabability of passing(0 to 1)
- Then we convert probability into class:
 - If probabaility > 0.5 - Pass
 - If probabaility < 0.5 - Fail

Hypothesis Function(Linear Part)

- First , Logestic Regression computes a linear combination of inputs: $z=wX +b$

where

w = weigth X =input feature b =bias z =linear output

Sigmoid Function

- To convert the linear output into a probabaility , Logistics Regression uses the Sigmoid Function.

Properties

- Output is always between 0 to 1
- Converts any real number into probability.
- Produces an S-shaped (Sigmoid) curve.

Probability integretation:

After applying the sigmoid function , we get probability:

Intepretation:

- if probability > 0.5 - class=1
- If probabaility < 0.5 - class=0

Decision Boundary

- The decision boundary is the threshold that seperates the two classes in logistic Regression.

Condiiton

- The decision boundary occurs when : $wX+b=0$

Where

- One side of the boundary = class 0
- Other side of the boundary = class 1

```
[6]: import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression

# Load dataset
df = pd.read_csv("student_data.csv")

# Features (X) and Target (y)
X = df[['Hours_Studied']]
y = df['Pass']
```

```
# Train-test split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Create model
model = LogisticRegression()

# Train model
model.fit(X_train, y_train)

# Predictions
y_pred = model.predict(X_test)

# Predict new student
new_student = pd.DataFrame({
    'Hours_Studied': [2]
})
prediction = model.predict(new_student)
print(y_pred)
print("Prediction (0=Fail, 1=Pass):", prediction[0])
```

```
[1 0]
Prediction (0=Fail, 1=Pass): 0
```

▼ K-Nearest Neighbours(KNN):

- K-Nearest Neighbors(KNN) is a supervised Machine Learning Algorithm used for :
 - Classification.
 - Regression.
- KNN is most commonly used for classification problems.
- The algorithm predicts the class of a new data type point by looking at the

Why is it called K-Nearest Neighbors?

K

- The number of nearest neighbors to consider

Nearest

- The closest data points based on distance

Neighbors

- The existing data points in the training dataset.

How KNN Works

- When a new data point arrives
 1. Calculate the distance between the new point and all training data points.
 2. Select the K nearest neighbors.
 3. Count the class labels of those neighbors.
 4. Assign the class that appears most frequently.

Distance Metric

- KNN identifies the nearest neighbors by calculating the distance between data points.
- the most commonly used distance metrics is euclidean distance.

where:

- d = distance between two points.
- (x_1, y_1) = First point
- (x_2, y_2) = Second point

Simple explanation

- Eucliden distance measures the

Interpretation

- The distance between point A and Point B is 5 units
- In KNN:
 - smaller distance = more similar points
 - larger distance = less similar points.

Why distance is important in KNN:

- KNN predicts a class by finding the nearest neighbors.
- The distance metric helps determine :
 - Which point are closest.
 - Which neighbors should participate in voting.

Choosing the Value of K

- The value of K plays a crutial role in the performance of the KNN algorithm.
- Choosing a very small or very large value of K can negatively affect prediction accuracy.

Case 1: Small K

Example

* $K = 1$

- The model looks at the only one nearest neighbor .

Advantage

- Captures local patterns effectively.
- Can model complex decision boundaries

Disadvantages

- Very sensitive to noise and outliers.
- A single incorrectly labeled point can lead to a wrong prediction .

Example

- Suppose the nearest neighbor is mistakenly as pass instead of fail.
- Since $K = 1$, the mode directly predicts Pass resulting

Case 2 : Large K

Example:

- $k = 10$
- The model considers many neighbors before making a prediction.

Advantages:

- More stable predictions.
- Less affected by noise data.

Disadvantages:

- Important local patterns may be ignored.
- Different classes may get mixed together.

Example:

- If most nearby students failed but many distant students passes , a large K may incorrectly predict Pass.

Visual Understanding

Small K	Large K
Sensitive to noise	Stable predictions
Complex decision boundary	Smooth decision boundary
Overfitting risk	Underfitting risk

Case 3 : Best Choice of K

A common rule of thumb is

where:

$K = \sqrt{N}$ N = Number of training samples.

Example:

Given:

* $N=100$

Curse of Dimensionality in KNN

What is the curse of dimensionality ?

- The curse of dimensionality refers to the problems that occur when the number of features (dimensions) in a dataset becomes very large.
- KNN relies on distance calculations to find the nearest neighbors. As the number of dimensions increases, distance calculations become less effective, making it difficult for KNN to identify truly similar data types.

What is dimension ?

Number of dimensions = 2

Example 2

Sample Dataset

Hours	Attendance	Age	Marks	Income	Salary
5	80	20	75	20000	30000

Number of dimensions = 6

What happens as dimensions increase ? ¶

As the number of features grows:

- data points become sparse (spread far apart).
- Distance calculations become less meaningful.
- Nearest neighbors become harder to identify.
- Model accuracy may decrease.

Why is it a Problem?

KNN works by finding the nearest neighbors using distance .

Low dimensions(2 or 3 features) Data points are relatively nearest neighbor are easily identify

Effects on KNN

1. Reduce Accuracy

- The identified nearest neighbors may not actually are similar

2. Increased Computation

- More features mean more distance calculations

3. Higher Memory Usage

- KNN stores the entire training dataset.

Solutions

Definiton

Feature selection

Dimensionality Reduction

Common tech

- * Principal Component Analysis(PCA)

* Linear Discriminant Analysis(LDA)

These techniques compress data into fewer dimensions.

What is decision Tree?

- A Decision Tree is a supervised machine learning algorithm used for :
 - Classification
 - Regression
- It works like a flowchart , where each decision leads to another question until a final prediction is reached.

Does a decision tree work ?

- A decision tree learns by asking a series of questions about the data.

At each step :

1. Select the best feature for splitting the data.
 2. Divide the data into smaller groups .
 3. Repeat the process on each group.
 4. Stop when a prediction can be made.
- This process is called recursive splitting.

Components of Decision Tree

1. Root Node:

- The top most node of the tree.

Example:

- Study hours > 5?
- This is the first question asked.

2. Decision Node

- A node where a decision or condition is checked.

Example:

* Attendance > 75?

3. Leaf Node

- The final output or prediction.

Example: ¶

Pass or fail

Gini Impurity

- Gini impurity is a measure used in decision trees to determine how mixed the classes are within a node.
- It tells us the probability of incorrectly classifying a randomly chosen sample if we assign a class label accordingly to the class distribution in that node.
- Think of a box containing students labeled as Pass or Fail.

Case 1 : Mixed Classes

- Think of a box containing students labeled as Pass or Fail.

Case 1 : Mixed Classes

Result Summary

Result	Count
Pass	5
Fail	5

- Total students = 10
- If randomly pick a student, it is difficult to predict whether the student belongs to the pass or fail class because both classes appear equally.

Case 2 : Pure classes

Result Summary

Result	Count
Pass	10
Fail	0